

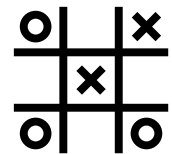
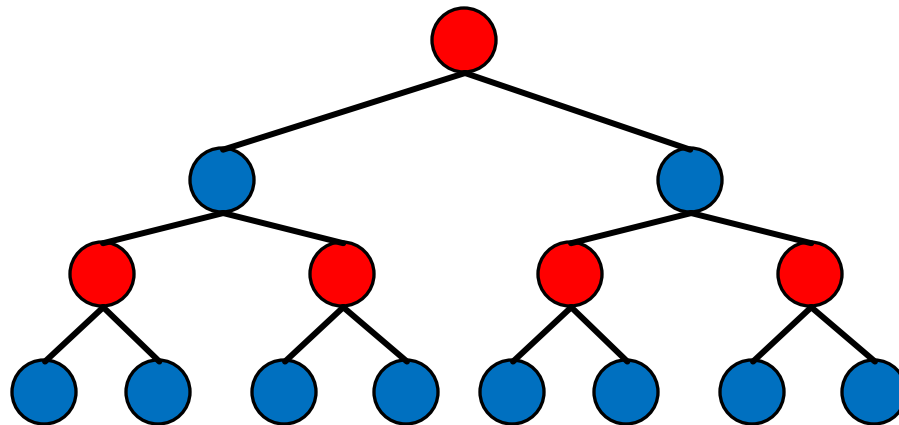
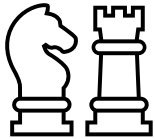


Nash Equilibria in Partial Information Games

AIMA Chapter 5.6

See also Osborne's "Introduction to Game Theory" Chapter 11

Subgame-perfect Nash equilibrium is our solution concept in adversarial search.



Nash equilibrium: the strategy of every player maximizes their utility given what other players are doing.

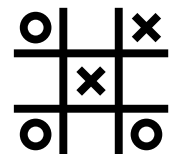
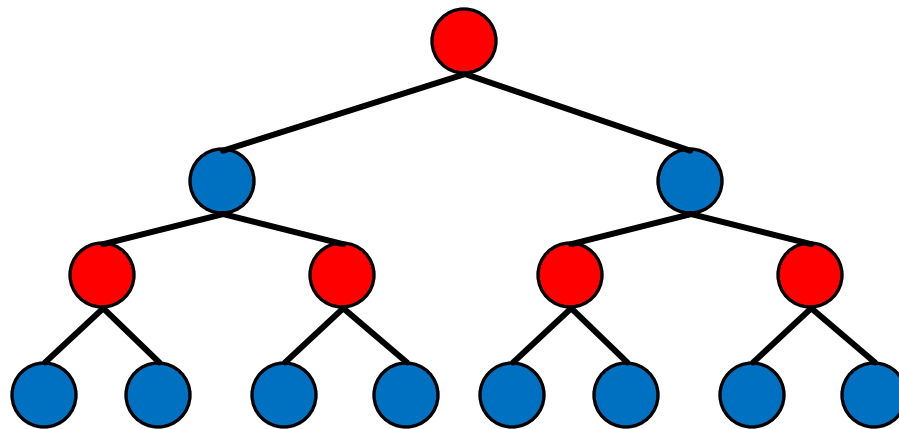
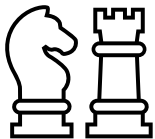
The strategy profile $(\vec{s}_1, \vec{s}_2)$ is a Nash equilibrium if for any other strategy $\vec{p}$ for player 1 and any other strategy $\vec{q}$ for player 2, we have

$$u_1(\vec{s}_1, \vec{s}_2) \geq u_1(\vec{p}, \vec{s}_2)$$

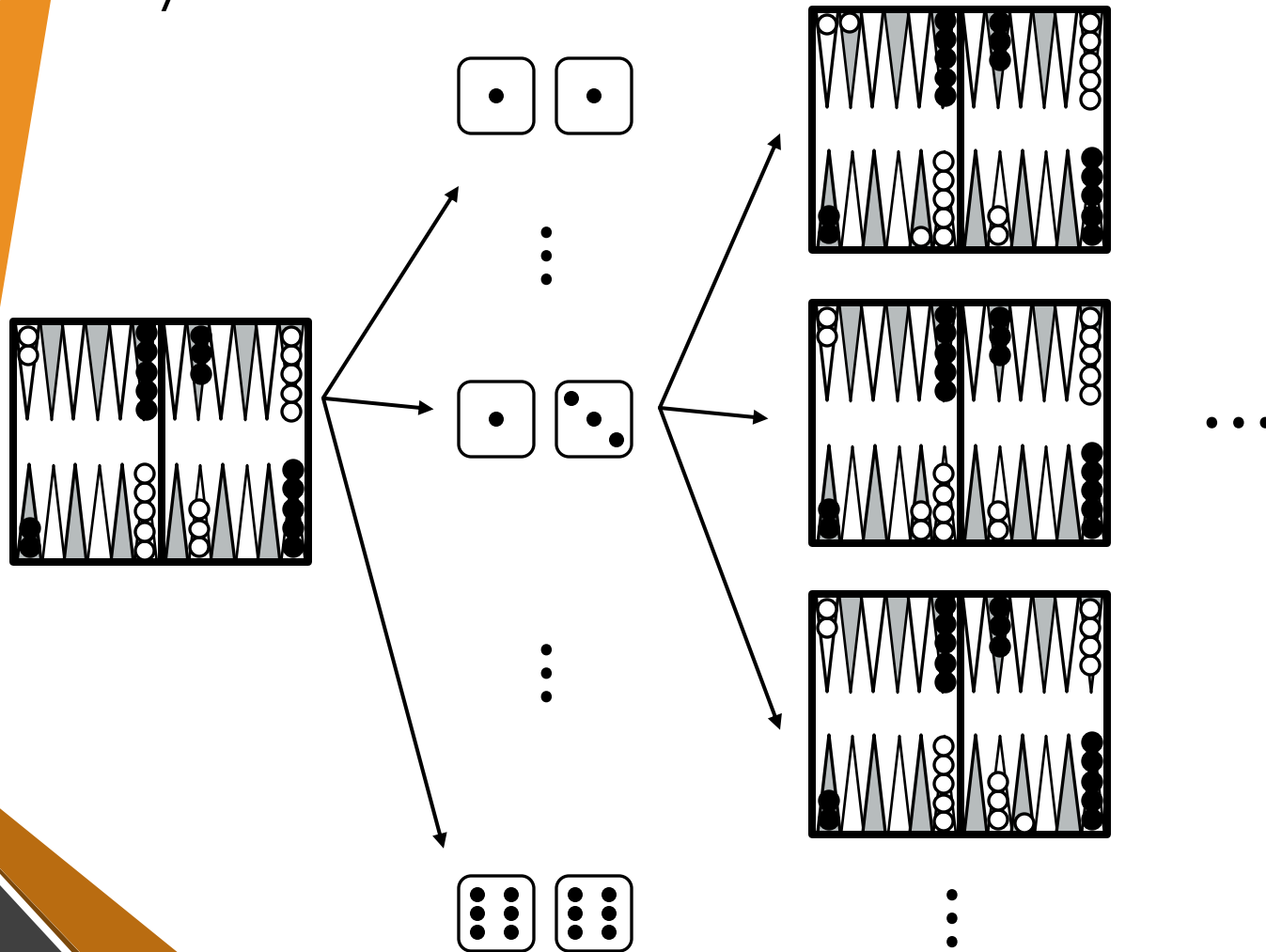
$$u_2(\vec{s}_1, \vec{s}_2) \geq u_2(\vec{s}_1, \vec{q})$$

Both players are best-responding to each other's strategy.

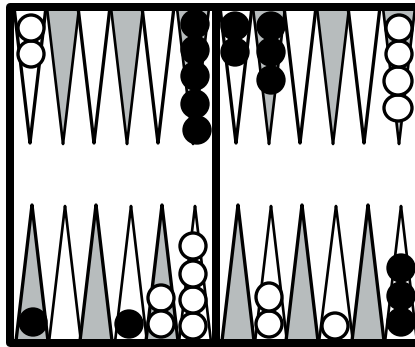
The strategy profile $(\vec{s}_1, \vec{s}_2)$ is a **subgame-perfect** Nash equilibrium if the definition holds for every subtree.



Randomness: some games have random moves. Modelled by *nature moves*.



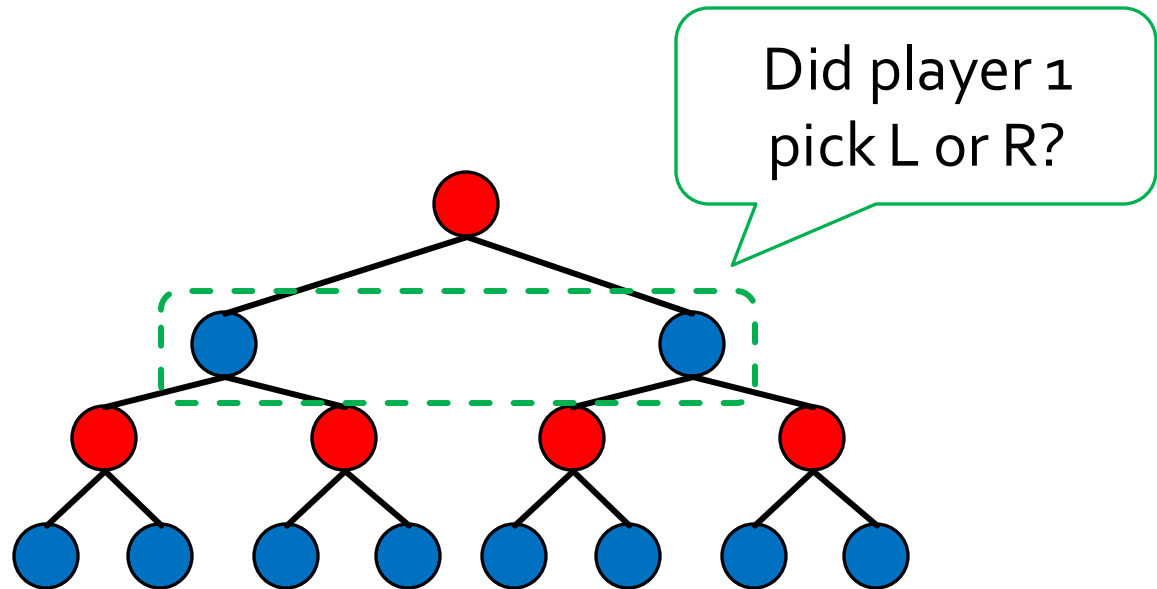
In backgammon, both players have complete knowledge of the current game state, i.e., the board.



The Nash equilibrium definition works just as well, taking expectations over the nature moves. For any player i and every strategy $\vec{q}$ for player i

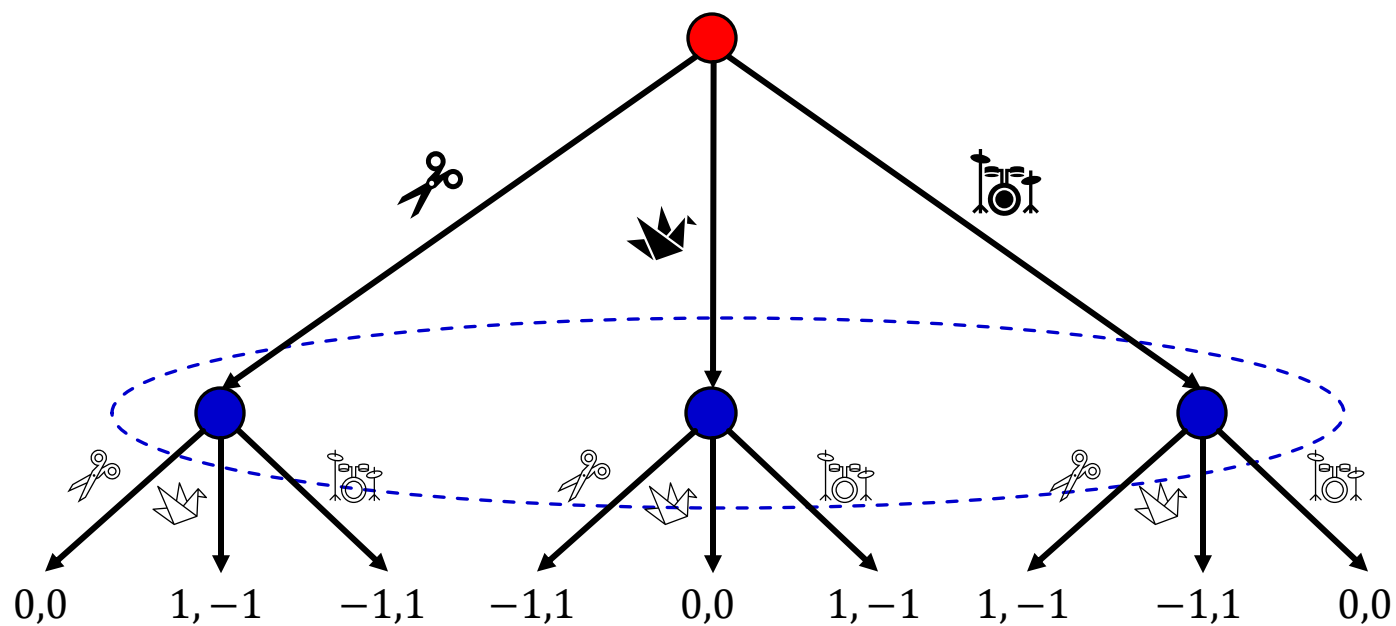
$$\mathbb{E}[u_i(\vec{s}_i, \vec{s}_{-i})] \geq \mathbb{E}[u_i(\vec{q}, \vec{s}_{-i})]$$

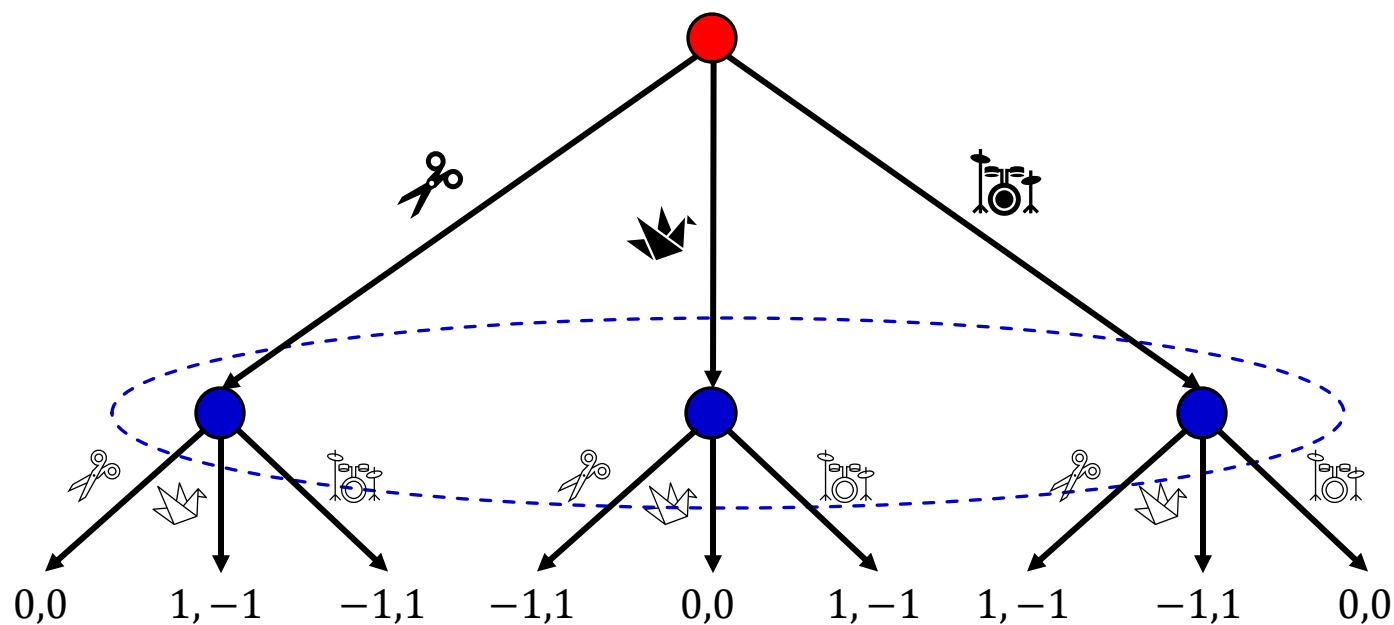
In some games, players don't have complete knowledge of the game state.



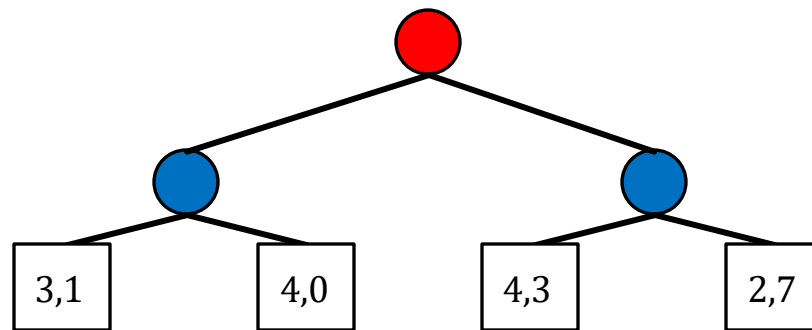


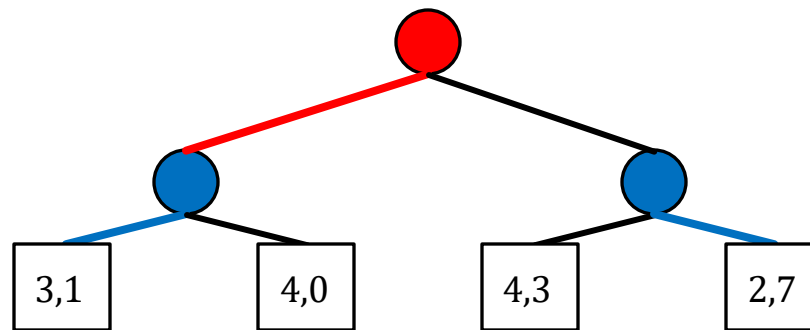
0,0	1, -1	-1,1
-1,1	0,0	1, -1
1, -1	-1,1	0,0

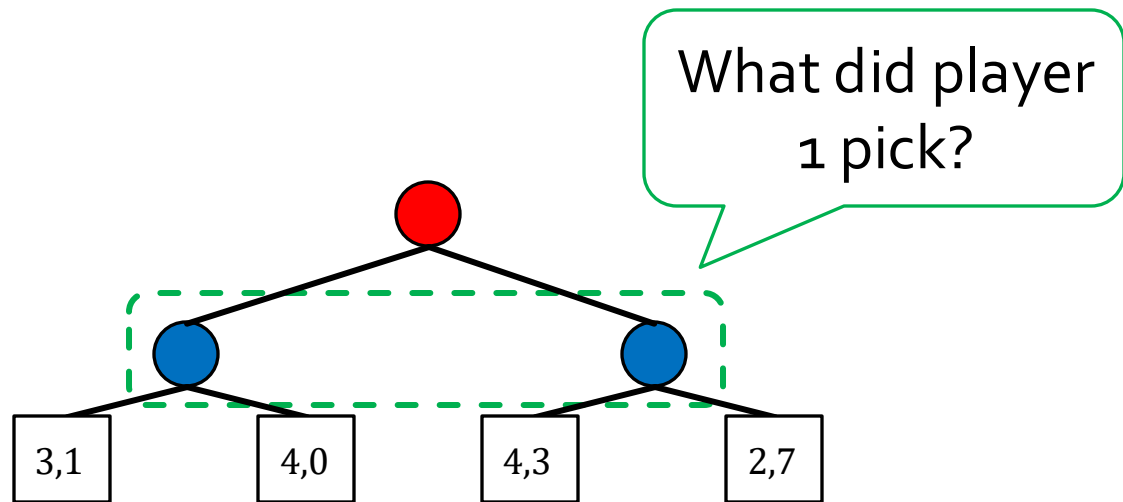


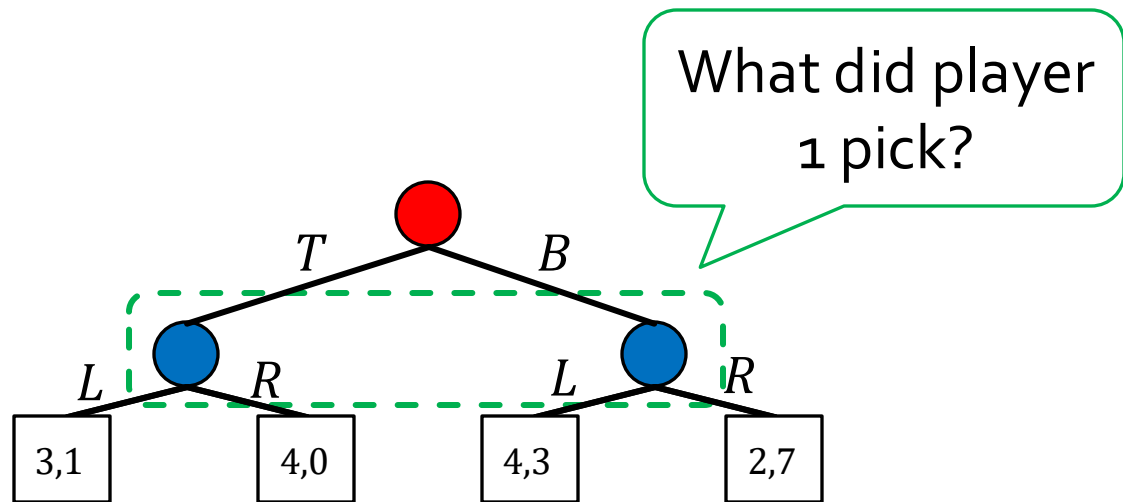


Are deterministic strategies Nash equilibria here?



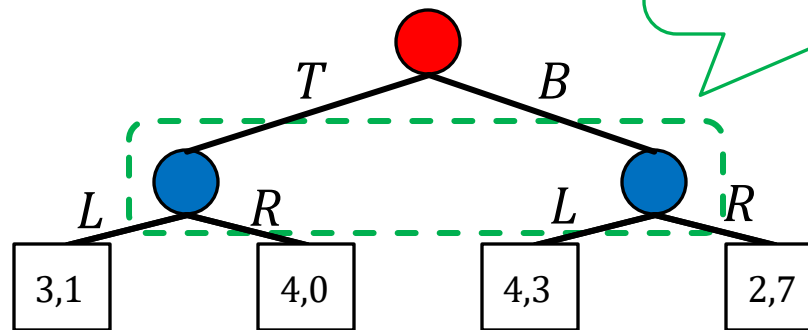






Should player 1 just always play T (or always play B)?

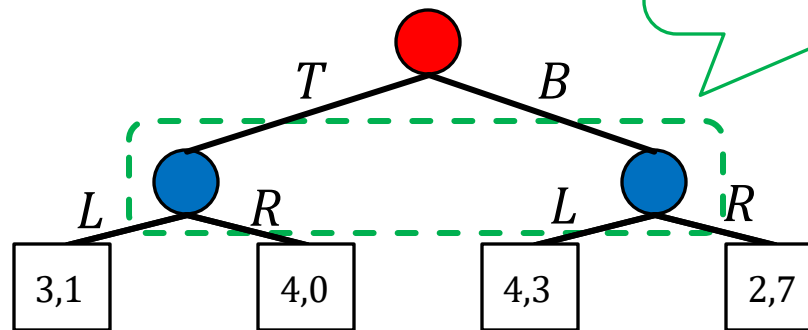
What did player 1 pick?



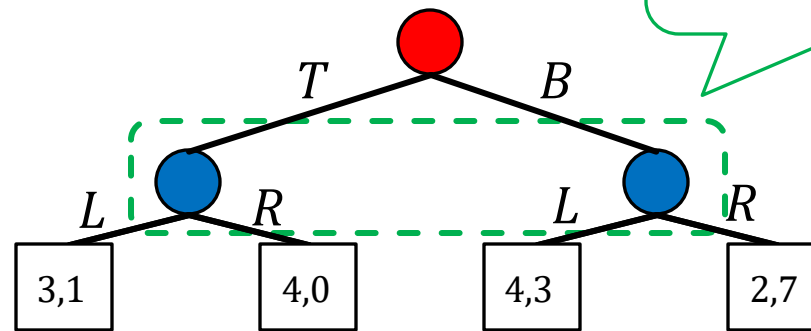
	L	R
T	3,1	4,0
B	4,3	2,7

Pause: is there a pure strategy Nash equilibrium here? 🐾🐾

What did player 1 pick?



	L	R
T	3,1	4,0
B	4,3	2,7



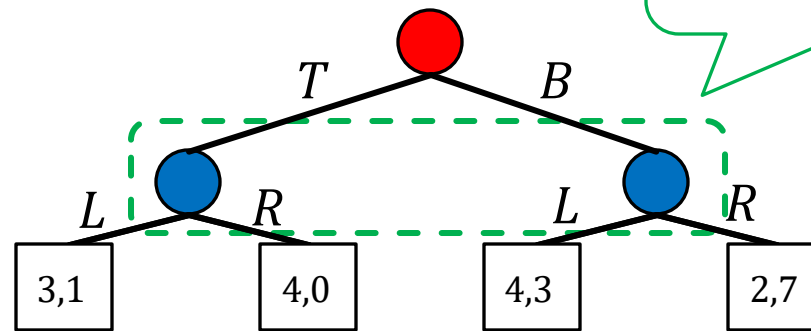
What did player 1 pick?



	<i>L</i>		<i>R</i>
<i>T</i>	3,1	←	4,0
<i>B</i>	4,3	→	2,7

Player 1's reasoning: I prefer *T* if player 2 plays *R*, but prefer *B* if player 2 plays *L*.

Player 2's reasoning: I prefer *L* if player 1 plays *T*, but prefer *R* if player 1 plays *B*.



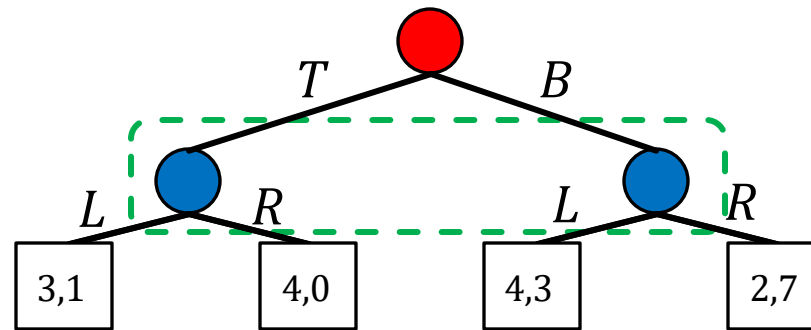
What did player 1 pick?



	<i>L</i>	<i>R</i>
<i>T</i>	3,1	4,0
<i>B</i>	4,3	2,7

Pure strategy profiles are not Nash equilibria.

What about randomized actions?



	<i>L</i>	<i>R</i>
<i>T</i>	3,1	4,0
<i>B</i>	4,3	2,7

Player 1: play T with prob. p and B with prob. $1 - p$.

Player 2: play L with prob. q and R with prob. $1 - q$.

Nash equilibrium concept is the same:

no player wants to change their (randomized) strategy given the other player's strategy.

$(\vec{p}, \vec{q})$ is a Nash equilibrium if

$u_1(\vec{p}, \vec{q}) \geq u_1(\vec{p}', \vec{q})$ for any other strategy $\vec{p}'$ that player 1 can pick.

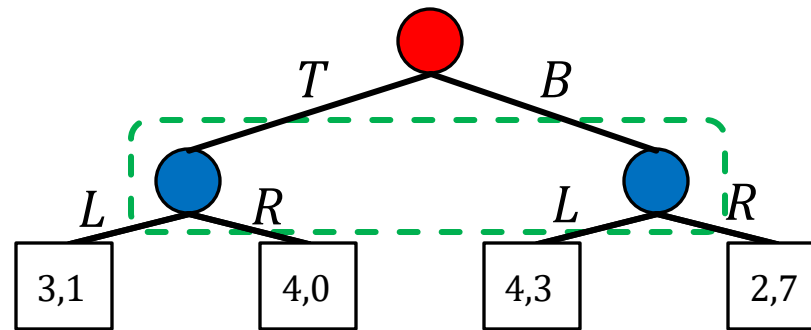
$u_2(\vec{p}, \vec{q}) \geq u_2(\vec{p}, \vec{q}')$ for any other strategy $\vec{q}'$ that player 2 can pick.

Observation: if $(\vec{p}, \vec{q})$ is a Nash equilibrium, then all actions played with positive probability give the player the same utility.

Proof: suppose that $(\vec{p}, \vec{q})$ is a Nash equilibrium, and that the action a gives strictly less utility than the action b .

$$u_1(a, \vec{q}) < u_1(b, \vec{q})$$

The strategy profile where b is played with probability $p_a + p_b$ and a is played with prob. 0 gives player 1 a strictly higher utility – then $\vec{p}$ does not maximize player 1's utility given $\vec{q}$, and $(\vec{p}, \vec{q})$ is **not** an equilibrium.



	<i>L</i>	<i>R</i>
<i>T</i>	3,1	4,0
<i>B</i>	4,3	2,7

Player 1: play T with prob. p and B with prob. $1 - p$.

Player 2: play L with prob. q and R with prob. $1 - q$.



	L	R
T	3,1	4,0
B	4,3	2,7

Player 1: play T with prob. p and B with prob. $1 - p$.

Player 2: play L with prob. q and R with prob. $1 - q$.

Suppose that (p, q) is a Nash equilibrium.

$$u_1(T, q) = 3 \cdot q + 4(1 - q) = 4 - q$$

$$u_1(B, q) = 4q + 2(1 - q) = 2 + 2q$$

	L	R
T	3,1	4,0
B	4,3	2,7

Suppose that (p, q) is a Nash equilibrium.

$$u_1(T, q) = 3 \cdot q + 4(1 - q) = 4 - q$$

$$u_1(B, q) = 4q + 2(1 - q) = 2 + 2q$$

$$4 - q \geq 2 + 2q \Leftrightarrow 2 \geq 3q \Leftrightarrow q \leq \frac{2}{3}$$

The **best response** of player 1 as a function of q :

$$BR_1(q) = \begin{cases} T & \text{if } q < \frac{2}{3} \\ B & \text{if } q > \frac{2}{3} \\ \text{anything} & \text{if } q = \frac{2}{3} \end{cases}$$

	L	R
T	3,1	4,0
B	4,3	2,7

Player 1: play L with prob. p and R with prob. $1 - p$.

Player 2: play A with prob. q and B with prob. $1 - q$.

Suppose that (p, q) is a Nash equilibrium.

$$u_2(p, L) = p + 3(1 - p) = 3 - 2p$$

$$u_2(p, R) = 0 \cdot p + 7(1 - p) = 7 - 7p$$

$$3 - 2p \geq 7 - 7p \Leftrightarrow 5p \geq 4 \Leftrightarrow p \geq \frac{4}{5}$$

	L	R
T	3,1	4,0
B	4,3	2,7

Suppose that (p, q) is a Nash equilibrium.

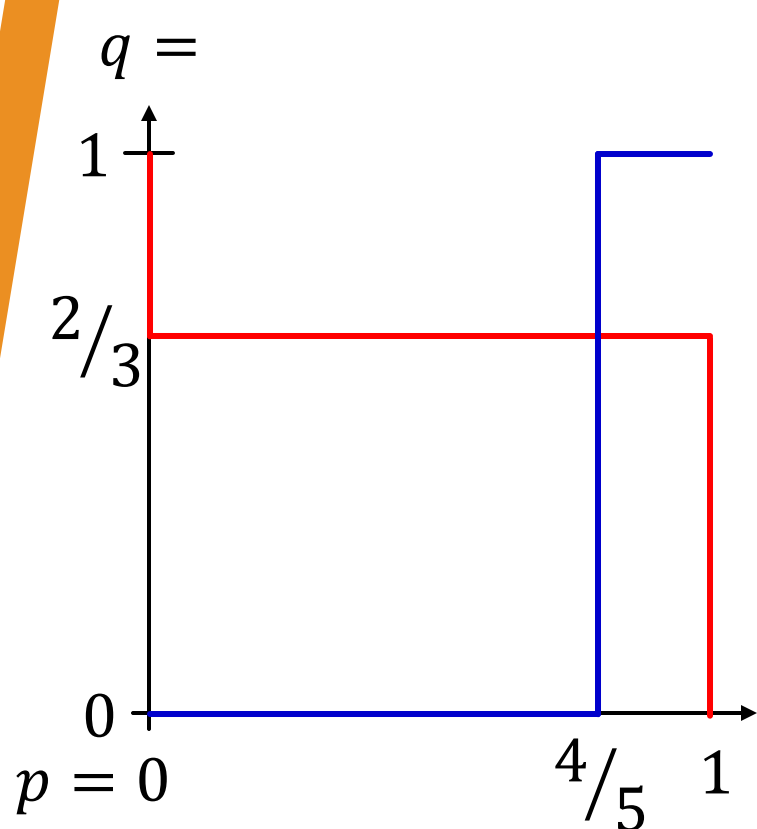
$$u_2(p, L) = p + 3(1 - p) = 3 - 2p$$

$$u_2(p, R) = 0 \cdot p + 5(1 - p) = 5 - 5p$$

$$3 - 2p \geq 5 - 5p \Leftrightarrow 5p \geq 2 \Leftrightarrow p \geq \frac{2}{5}$$

The **best response** of player 2 as a function of p :

$$BR_2(p) = \begin{cases} L & \text{if } p > \frac{2}{5} \\ R & \text{if } p < \frac{2}{5} \\ \text{anything} & \text{if } p = \frac{2}{5} \end{cases}$$



$$BR_1(q) = \begin{cases} T & \text{if } q < \frac{2}{3} \\ B & \text{if } q > \frac{2}{3} \\ \text{anything} & \text{if } q = \frac{2}{3} \end{cases}$$

$$BR_2(p) = \begin{cases} L & \text{if } p > \frac{4}{5} \\ R & \text{if } p < \frac{4}{5} \\ \text{anything} & \text{if } p = \frac{4}{5} \end{cases}$$

The only case where the players are best-responding to each other is when $p = \frac{4}{5}, q = \frac{2}{3}$. This is the Nash equilibrium.